

AKOLA, India

WMO: 429330

Lat: 20.700N

Lon: 77.033E

Elev: 282

StdP: 97.98

Time Zone: 5.50 (IND)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.9	(c) 13.3	(d) -0.9	(e) 3.6	(f) 30.6	(g) 1.2	(h) 4.3	(i) 28.2	(j) 3.3	(k) 27.0	(l) 2.5	(m) 26.6	(n) 0.1	(o) 140	(p) 0.322

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.1	43.5	22.0	42.5	21.7	41.3	21.6	27.0	32.7	26.5	31.8	26.1	31.1	1.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.7	21.7	28.9	25.3	21.2	28.4	25.0	20.8	28.1	87.3	32.8	85.2	31.9	83.4	31.1	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.2	(b) 3.3	(c) 2.8	(e) 9.5	(f) 45.6	(g) 1.1	(h) 0.9	(i) 8.6	(j) 46.2	(k) 8.0	(l) 46.7	(m) 7.3	(n) 47.2	(o) 6.5	(p) 47.8		
(a) 4.2	(b) 3.3	(c) 2.8	(e) 6.5	(f) 28.6	(g) 1.3	(h) 0.9	(i) 5.6	(j) 29.2	(k) 4.9	(l) 29.7	(m) 4.2	(n) 30.1	(o) 3.3	(p) 30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	22.5	25.6	29.4	33.8	36.0	32.2	28.1	27.4	28.1	27.8	25.5	22.7
	DBStd	4.44	2.11	2.32	2.43	1.93	1.68	2.97	1.87	1.41	1.33	1.62	1.73	2.08
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6668	387	437	603	714	805	666	562	541	542	552	466	394
	CDD18.3	3627	129	203	344	464	547	416	303	282	292	293	216	136
	CDH23.3	46647	1224	2240	4681	7531	9339	6144	3312	2809	3020	3121	2012	1214
	CDH26.7	26556	444	1164	2886	5225	6864	3843	1248	881	1147	1527	903	424
Wind	WSAvg	0.8	0.4	0.6	0.6	0.9	1.5	1.4	1.1	1.0	0.8	0.4	0.3	0.3
Precipitation	PrecAvg	785	11	7	13	5	10	134	216	185	127	46	16	9
	PrecMax	1278	58	51	144	63	58	322	440	412	328	166	132	123
	PrecMin	388	0	0	0	0	0	7	35	0	0	0	0	0
	PrecStd	185	17	12	24	10	16	81	98	89	79	49	31	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	37.2	41.6	44.5	45.0	43.4	36.6	33.8	35.5	36.5	34.6	32.9
		MCWB	18.3	20.2	19.9	21.1	22.2	23.8	24.5	25.2	24.6	22.1	20.9	19.8
	2%	DB	31.8	35.8	40.1	43.0	43.9	41.4	34.7	32.6	34.0	35.4	33.3	31.4
		MCWB	18.4	19.3	19.4	21.0	22.3	23.6	25.0	25.1	24.9	22.1	20.8	19.4
	5%	DB	30.6	34.4	38.8	42.0	43.0	39.7	33.1	31.7	32.8	34.4	32.4	30.3
		MCWB	18.0	19.1	19.5	20.5	22.1	24.0	25.1	25.1	25.2	22.2	20.6	18.9
	10%	DB	29.3	33.0	37.4	40.9	41.9	38.1	31.8	30.7	31.7	33.3	31.5	29.2
		MCWB	17.7	18.6	19.2	20.6	22.1	24.2	25.1	25.0	25.2	22.2	20.4	18.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.0	23.5	23.7	26.0	28.1	27.5	27.1	27.2	27.2	26.4	25.3	22.9
		MCDB	26.9	33.3	34.8	41.3	38.0	33.3	31.6	30.8	31.5	31.6	30.7	29.6
	2%	WB	20.2	21.9	22.4	23.5	26.5	26.8	26.5	26.5	26.6	25.8	24.5	21.6
		MCDB	27.8	32.1	33.4	36.9	36.7	33.5	30.7	30.0	30.8	30.4	29.6	28.3
	5%	WB	19.5	20.8	21.5	22.5	25.2	26.3	26.1	26.0	26.2	25.2	23.6	20.7
		MCDB	27.6	30.2	32.8	36.5	35.7	33.0	30.1	29.5	30.3	29.5	28.3	27.7
	10%	WB	18.6	19.8	20.7	21.7	24.4	25.9	25.7	25.6	25.8	24.7	22.7	19.7
		MCDB	27.2	29.8	32.5	36.4	36.0	32.6	29.4	29.0	29.7	28.7	27.3	27.0
Mean Daily Temperature Range	5% DB	MDBR	14.5	14.7	15.2	14.5	12.1	9.4	6.0	5.8	7.6	11.9	13.4	14.6
		MCDBR	15.9	16.2	17.1	15.6	13.0	11.3	8.4	7.7	9.5	13.6	14.2	15.2
		MCWBR	6.2	6.1	5.6	5.0	3.8	3.1	2.2	2.3	2.9	3.9	4.7	6.1
	5% WB	MCDBR	12.9	13.7	14.3	13.5	11.4	9.5	6.6	6.4	7.8	9.1	9.7	12.5
		MCWBR	5.7	6.5	5.7	5.0	4.5	3.0	2.5	2.6	3.0	3.4	3.8	5.8
Clear-Sky Solar Irradiance	taub	0.471	0.485	0.510	0.544	0.569	0.622	0.590	0.557	0.544	0.527	0.514	0.470	
	taud	2.010	1.997	1.960	1.928	1.909	1.838	1.918	2.011	2.026	2.005	1.971	2.026	
	Ebn at Noon	790	808	805	782	755	710	734	762	767	760	743	774	
	Edh at Noon	162	174	187	196	198	210	195	178	173	170	168	155	
All-Sky Solar Radiation	RadAvg	4.64	5.51	6.37	6.97	7.12	5.33	3.87	3.89	4.80	5.21	4.70	4.47	
	RadStd	0.18	0.23	0.30	0.20	0.23	0.54	0.45	0.38	0.35	0.27	0.27	0.14	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	+0.45	+1.05	N/A	+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page